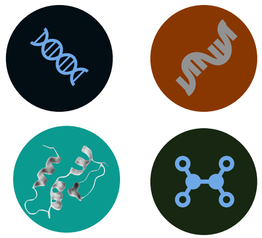
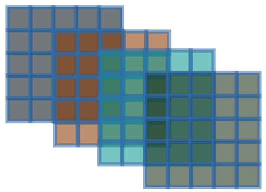
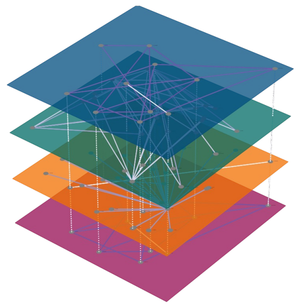




Basics of Downstream Proteomics Analysis

3rd June 2026

Alberto Santos

The Novo Nordisk Foundation Center for Biosustainability

Multi-omics Network Analytics (MoNA)

DTU Health Tech
Department of Health Technology

Multimodal Data

Implementing tools to process, integrate, and analyse multimodal data. Diving into the benefits of harmonising multimodal data that converge to provide a comprehensive view of complex biological systems. Specifically we are interested in high-throughput multi-omics data generated using Mass spectrometry technology (proteomics and metabolomics) and metaomics data (metagenomics and metaproteomics).

Knowledge Graphs

Building High-quality Knowledge Graphs. Using and developing Knowledge Graph technologies and methods to structured data and to connect them to existing biological knowledge. These structures facilitate analysis and interpretation of complex data. We are contributing to a groundbreaking field by developing tools and methods to build, assess and investigate Knowledge Graphs and applying them to solve challenges in biology and health.

Graph Machine Learning

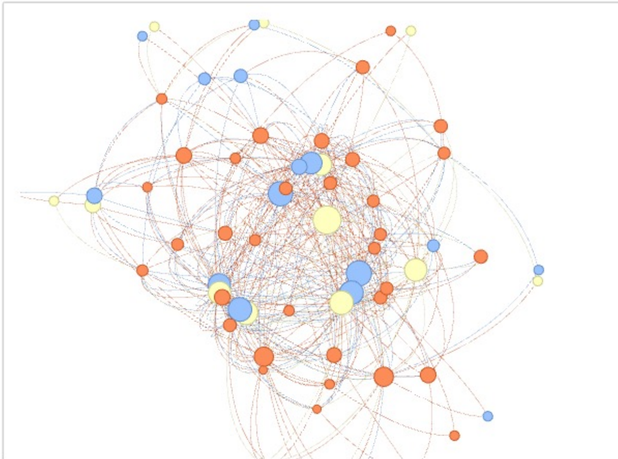
Developing and Applying Novel Methods on Graphs. Unleashing the power of Machine Learning on Graphs, a cutting-edge approach to extracting valuable insights from network data. We explore how this fusion of machine learning and graph theory helps to recognize patterns, generate predictions, and discovering new knowledge across a multitude of applications, including biological and medical networks.

Open Science

Data Science Democratisation. Focusing on data literacy training as a means to reduce inequality, and promoting open science by making all research, data content, and software open and accessible.

Microbial Communities

Exploring Microbial Communities and their Environments. Integrating multiple biological resources to unravel the assembly, interaction and adaptation mechanisms of microbial networks, offering insights into their functions and inpact on ecosystems, and how changes affect those communities.

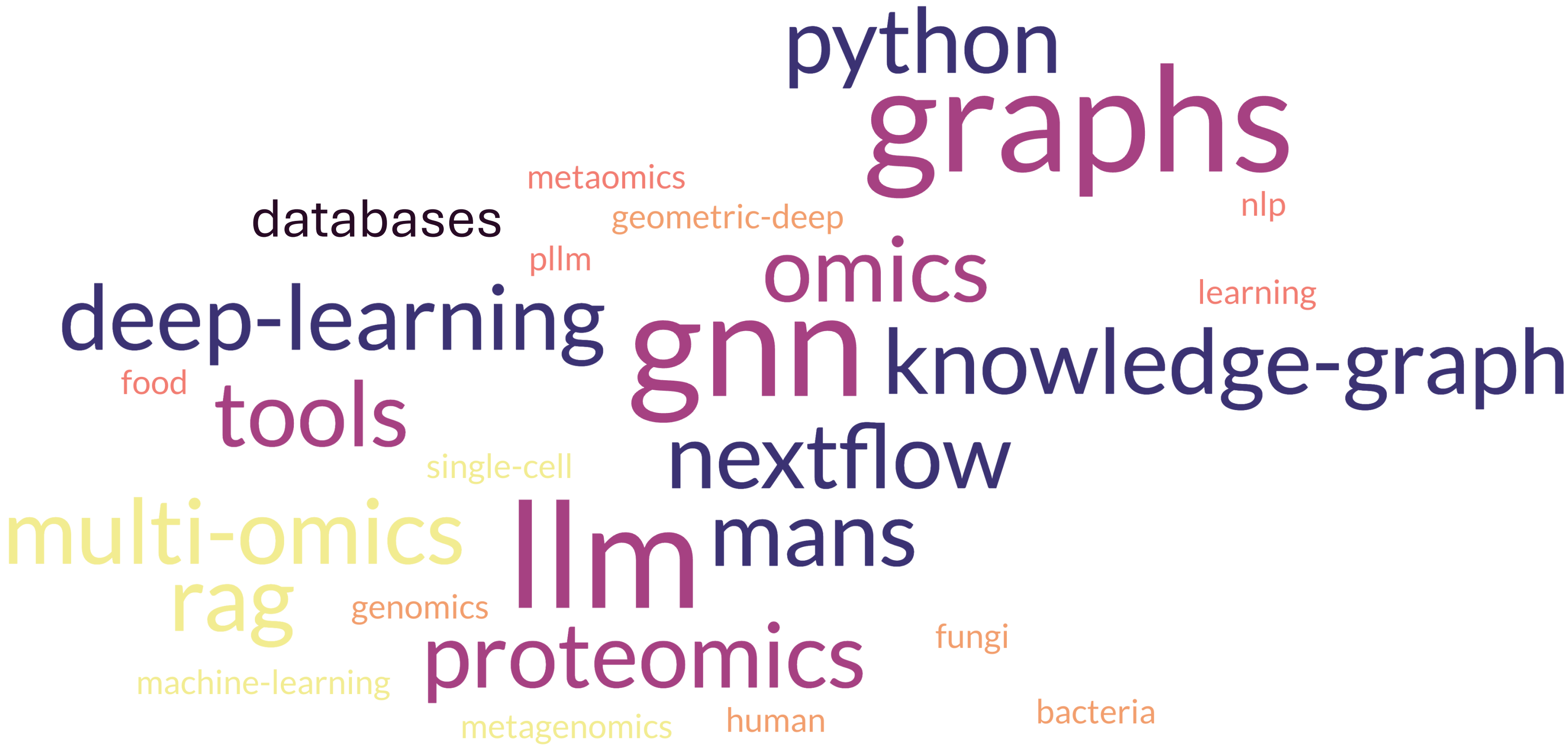


Clinical Computational Omics

Providing tools for the analysis and interpretation of clinical omics data. Integration of high-throughput omics data with computational and bioinformatics approaches to advance precision medicine and disease research. These projects aim to identify biomarkers, uncover disease mechanisms, and tailor treatments based on individual molecular profiles.

https://multiomics-analytics-group.github.io/

Research


python
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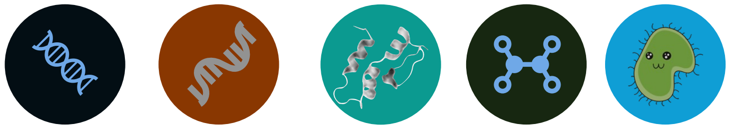
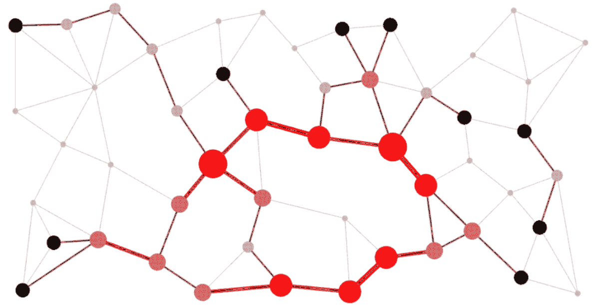
human

Graphs

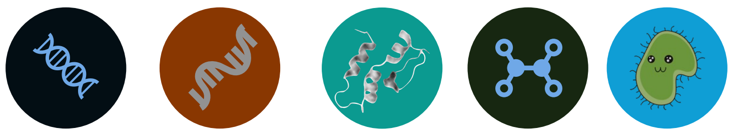




Why Omics?

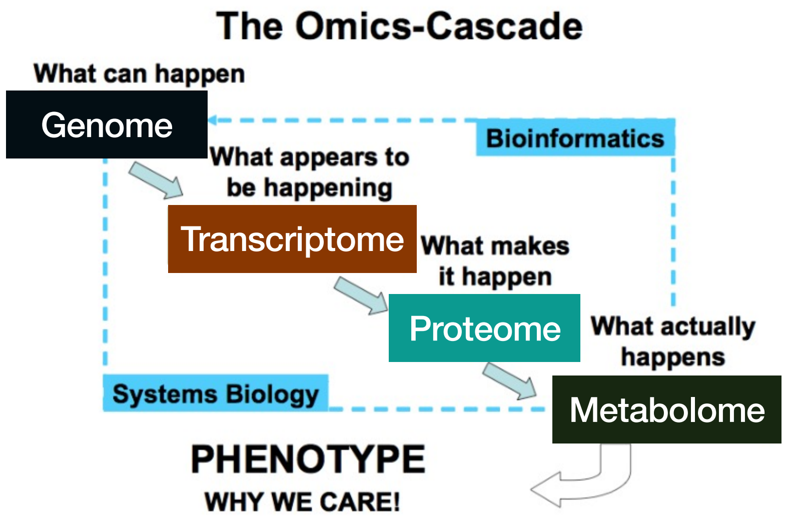
Understanding biology on a large scale

- Fields of study that aim to **map**, **quantify**, and **understand** sets of biological molecules within an organism or system— genes, proteins, metabolites, and more
- Provide:
  - **Holistic View** beyond single-gene or single-protein studies, providing a comprehensive view of biological processes
  - **High-Resolution Data** generated with high-throughput technologies

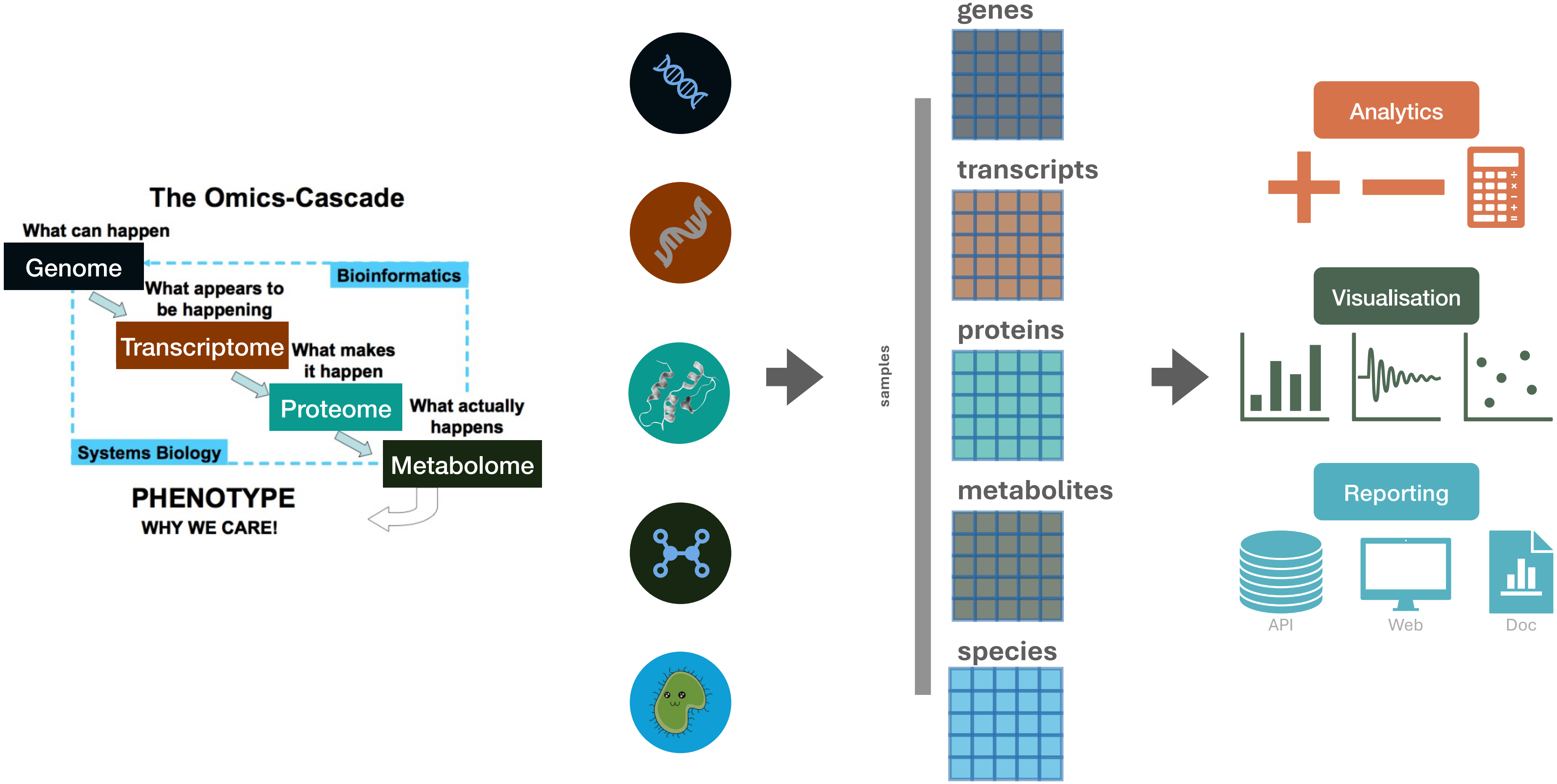


Types of Omics

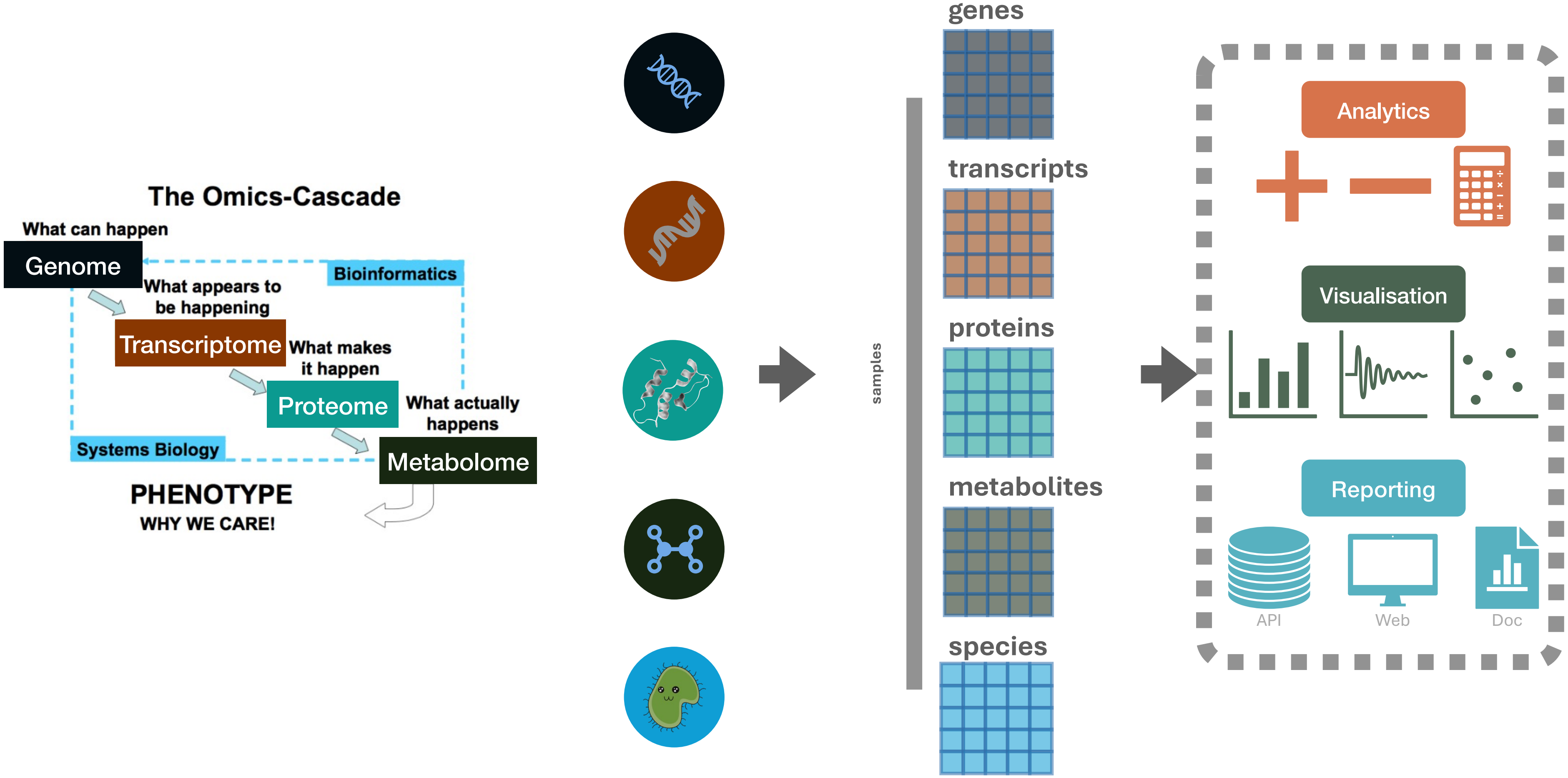
- **Genomics** Study of the genome, which includes all DNA within an organism
  - Sequence, structure, and function of genes
  - Key technology — Next-generation sequencing (NGS)
- **Transcriptomics** Study of the transcriptome, which is the complete set of RNA transcripts
  - Gene expression and regulation
  - Key technology — RNA sequencing (RNA-seq)
- **Proteomics** Study of the proteome, or the complete set of proteins in a cell or organism
  - Protein structure, function, interactions, and modifications
  - Key technology — Mass spectrometry (MS)
- **Metabolomics** Study of the metabolome, which includes all small-molecule metabolites in a cell or biological system
  - Cellular processes and metabolic pathways
  - Key technology — Mass spectrometry (MS)
- **Metaomics** Studies the collective genetic material, proteins, metabolites, and other molecular components from entire communities of organisms in a specific environment, without needing to isolate or culture individual species.



Downstream Analysis



Downstream Omics Analysis



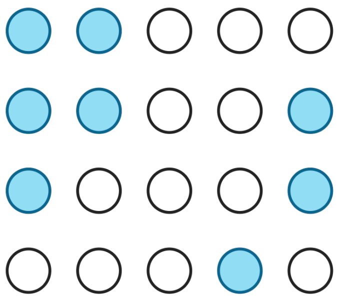
Downstream Proteomics Analysis

Objective

Main goals
- Identify **significant changes**
- Infer **biological meaning**
- **Integrate** with other omics data

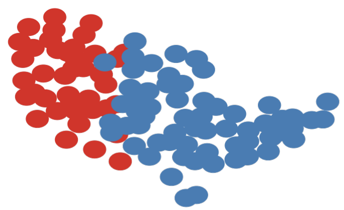
Challenges
- High dimensionality, small sample sizes
- Missing values and batch effects
- Interpretation bias in functional analysis
- Reproducibility

Missing Values



High-dimensionality

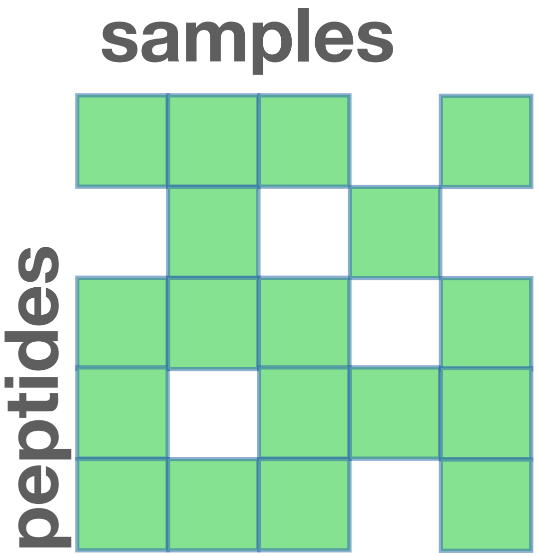
Batch Effects / Reproducibility



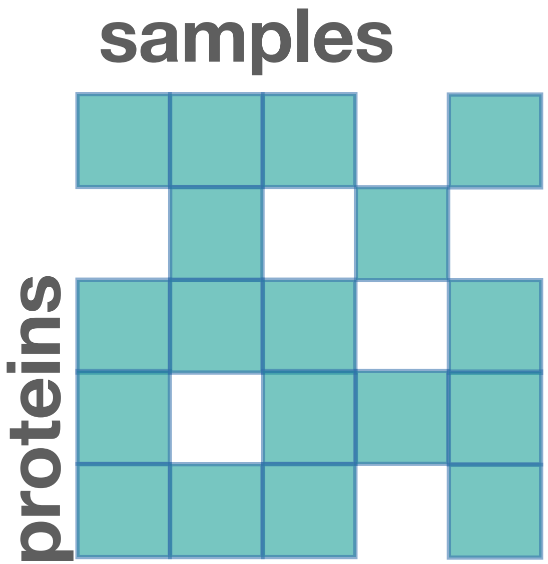
The Starting Point

Sample Metadata

Peptide Matrix



Protein Matrix



Sample Metadata Standard: SDRF

Sample and Data Relationship Format (SDRF) is a standardized, machine-readable metadata framework that directly connects biological samples to their mass spectrometry raw files. ensures experimental reproducibility and automated reuse.

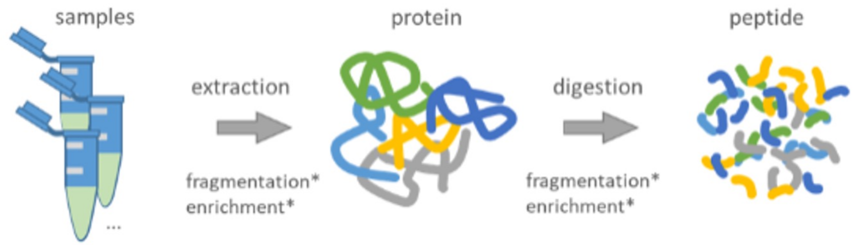
Column Prefix	Standard Field Examples	Description / Usage
characteristics[...]	organism, tissue, disease, age	Biological metadata describing the sample source.
comment[...]	instrument, fraction, data file	Technical metadata regarding data acquisition and files.
factor value[...]	treatment, genotype, timepoint	Variables used for **statistical grouping** and comparison.
label	TMT126, SILAC heavy, label-free	Quantification strategy used for the specific sample.

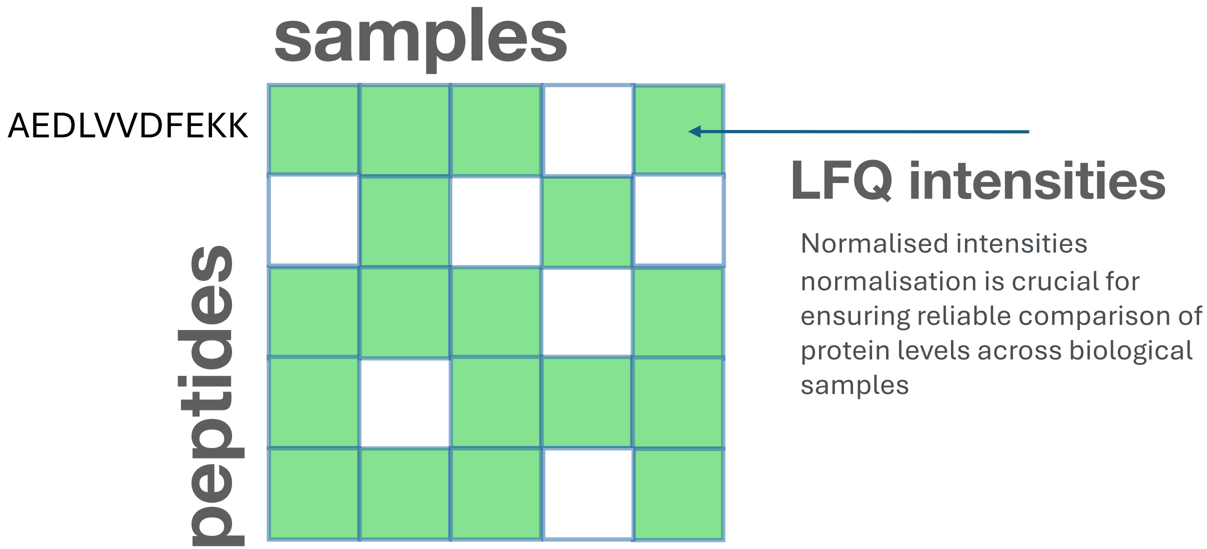
Benefit

It allows **automated pipelines** to recognize sample groups and specific metadata needed to execute statistical comparisons and analyses

https://www.ebi.ac.uk/pride/markdownpage/sdrf

Peptide Matrix



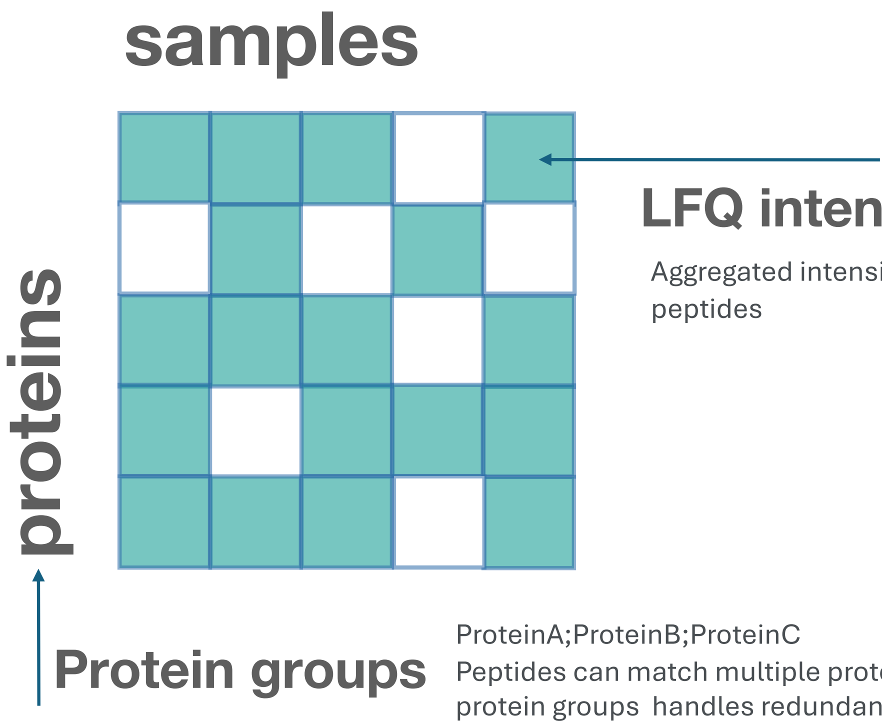


Raw intensity measures the total signal a mass spectrometer detects for a protein, **LFQ intensities** are normalised intensities to make them comparable across samples.

Multiple peptide sequences can map back to the same parent protein ID.

These peptide intensities are **statistically aggregated** into a single protein-level intensity.

Protein Matrix



LFQ intensities

Aggregated intensities from peptides

Protein groups

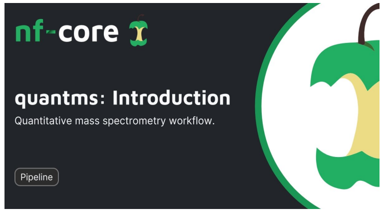
ProteinA;ProteinB;ProteinC
Peptides can match multiple proteins; protein groups handles redundancy in the matching of peptides to protein hits.
Leading protein -> first reported protein

A **protein group** is a collection of proteins that cannot be distinguished from one another based on the identified peptide evidence.

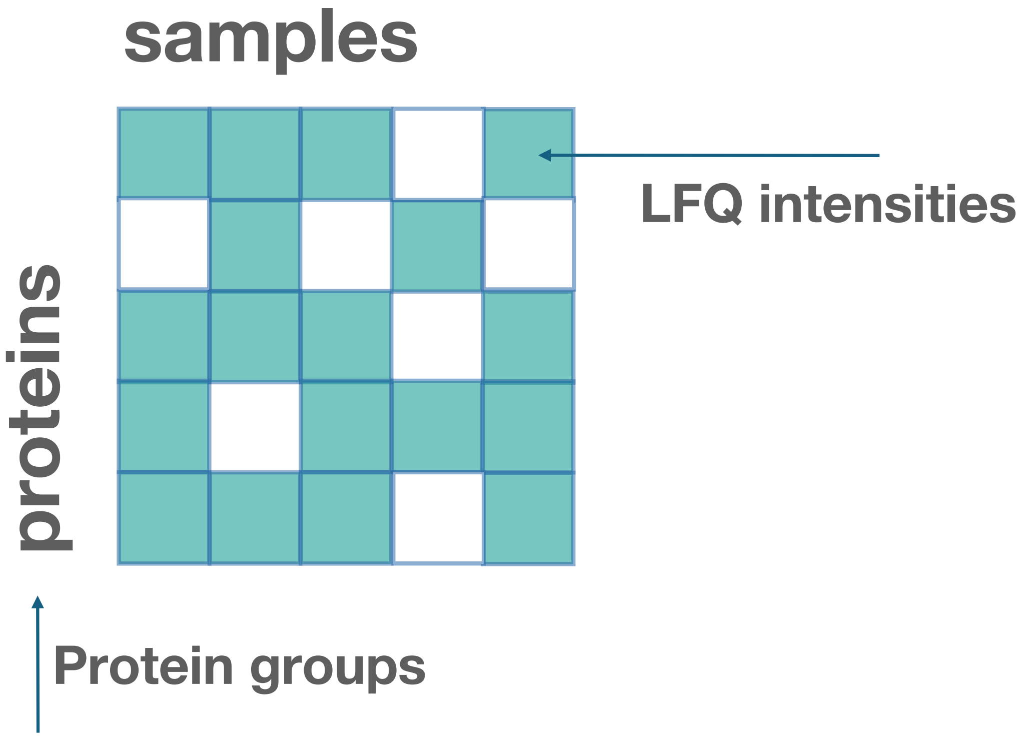
How are protein groups selected? Algorithms find the **minimum number of proteins** required to explain all the observed peptides

e.g., if Protein A is identified by 3 unique peptides, it gets its own independent group.
Rule 2: If Protein B and Protein C share 4 peptides, but Protein B also has 1 unique peptide, the algorithm selects Protein B.

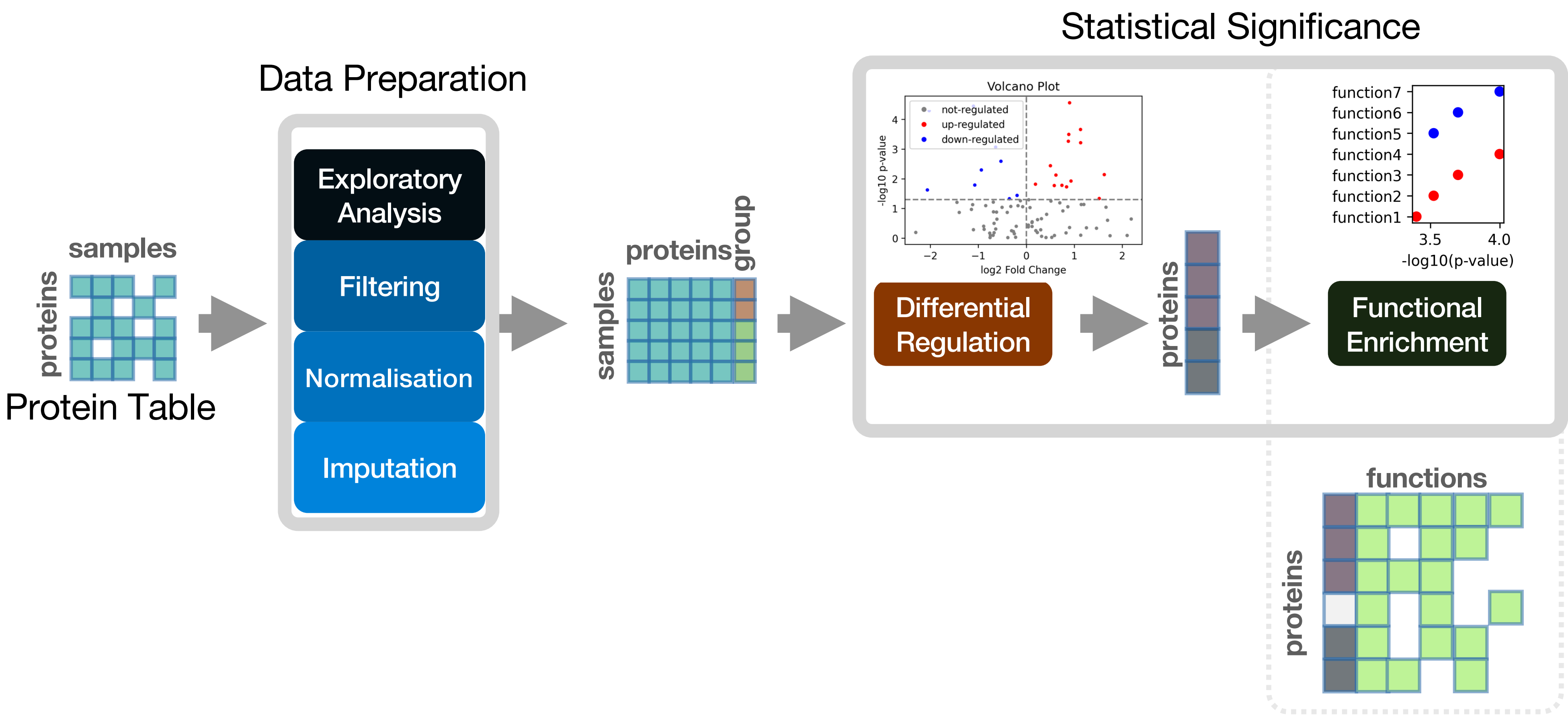
Proteomics Data



Spectronaut®
powered by Pulsar



Analytical Workflow



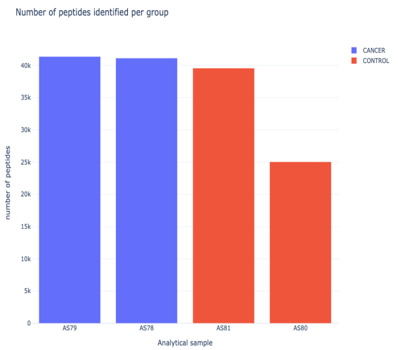
Data Preparation

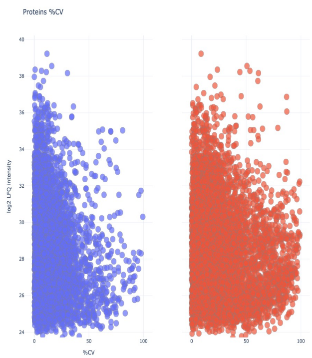
Exploratory Data Analysis

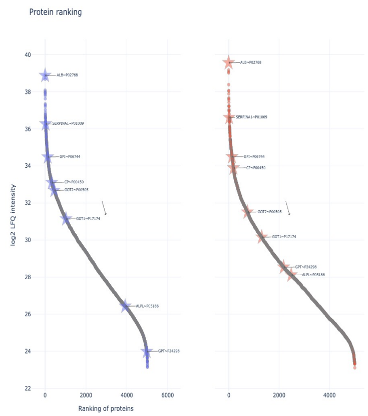
Exploratory Analysis

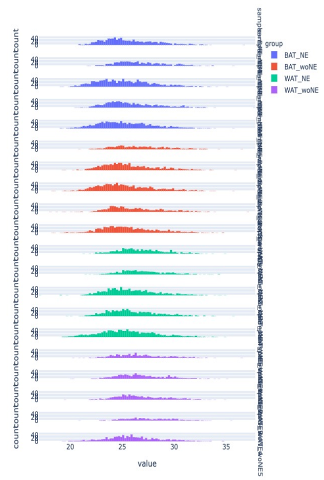
Filtering

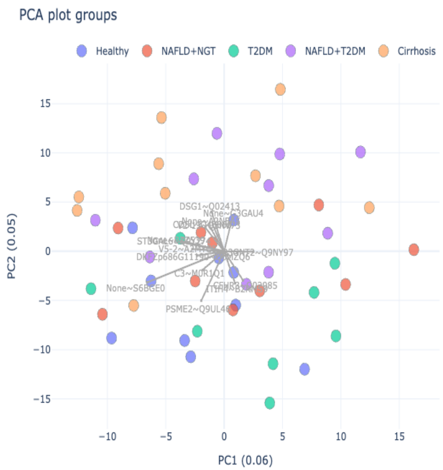
- **Exploratory Analysis**: understand the structure and quality of the data (peptide/protein number, intensities, dispersion)
- **QC/Filtering**: remove proteins and samples that do not meet quality criteria (e.g., missing too many values)





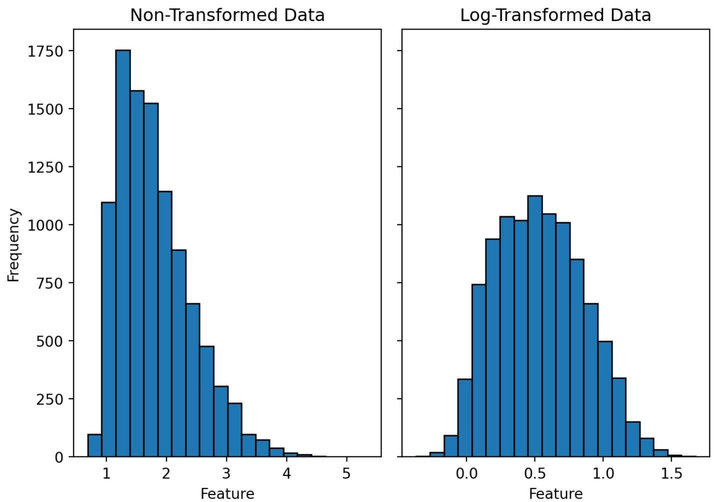






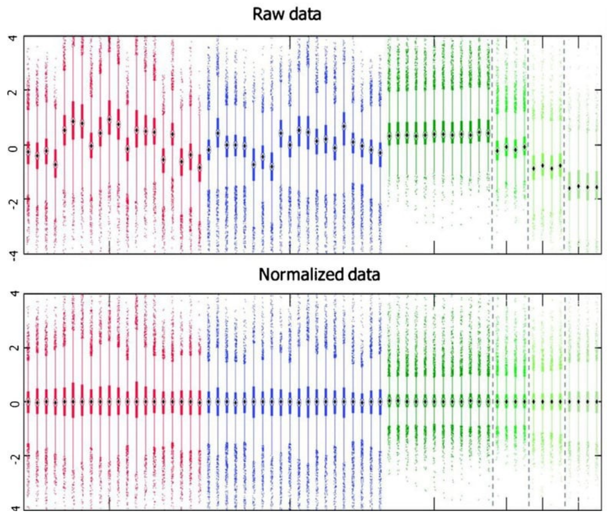
Log2 Transformation

Protein abundances span orders of magnitude. Log-transformation stabilizes variance and approximates a **Normal Distribution**, a key requirement for parametric tests.



Normalization

Corrects technical biases (e.g., loading differences). Methods like **Median** or **Quantile** normalization ensure intensities are globally aligned across all samples.



Handling Missing Values

MAR

Missing at Random: Technical errors or stochastic detection. Best handled by **K-Nearest Neighbors (KNN)** imputation.

MNAR

Missing Not at Random: Protein is below the detection limit. Fixed via **Left-Censored (Downshifted)** Normal Imputation.

Filtering

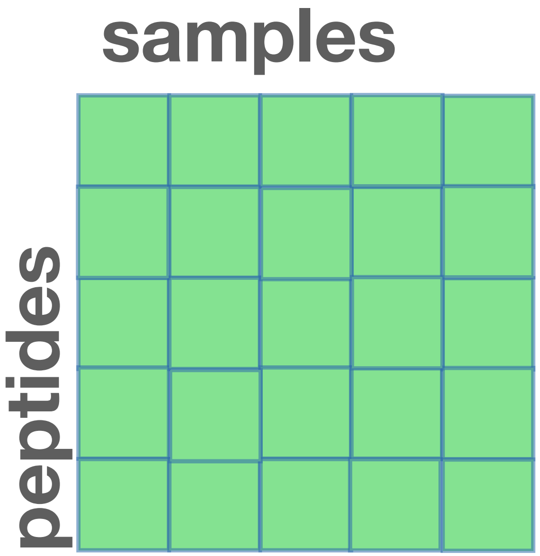
Removing low-quality proteins (e.g., missing in >50% of replicates) before imputation to reduce statistical noise.

Statistical Significance

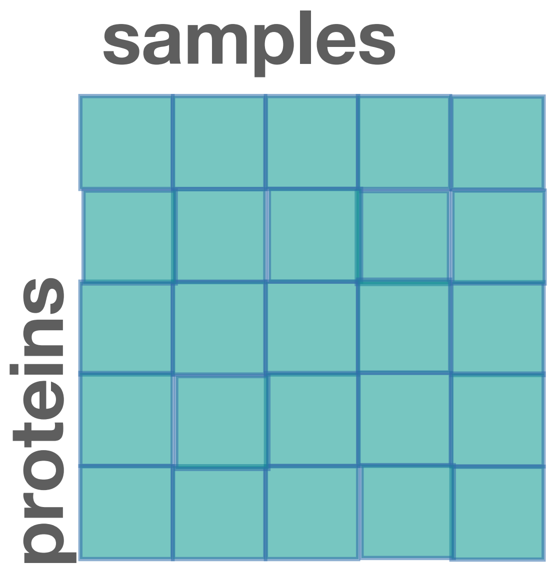
The Starting Point

Sample Metadata

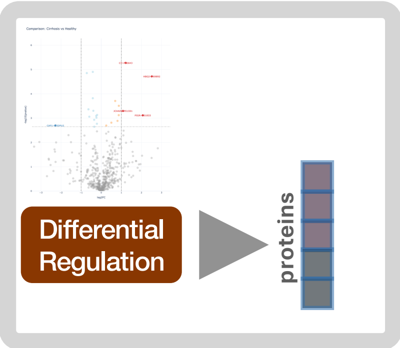
Peptide Matrix

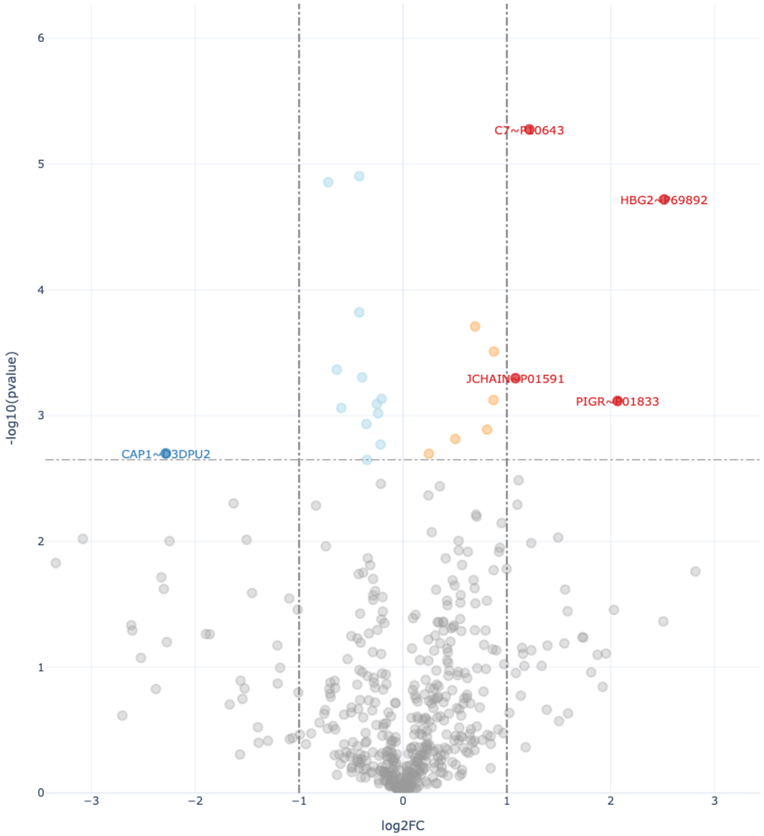


Protein Matrix



Data Analysis



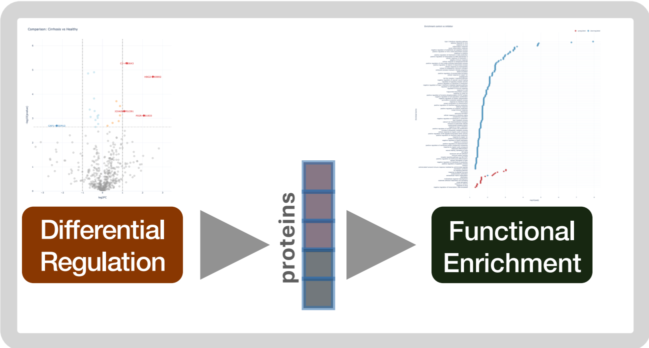


Volcano Plot

Differential Regulation: Apply appropriate statistical tests to compare protein intensities between groups — T-test, ANOVA

Multiple test correction (e.g., Benjamini-Hochberg False Discovery Rate (FDR)). If you test 5,000 proteins with $p < 0.05$, you expect **250 false positives** by chance. We control this using the **FDR**.

Functional Enrichment

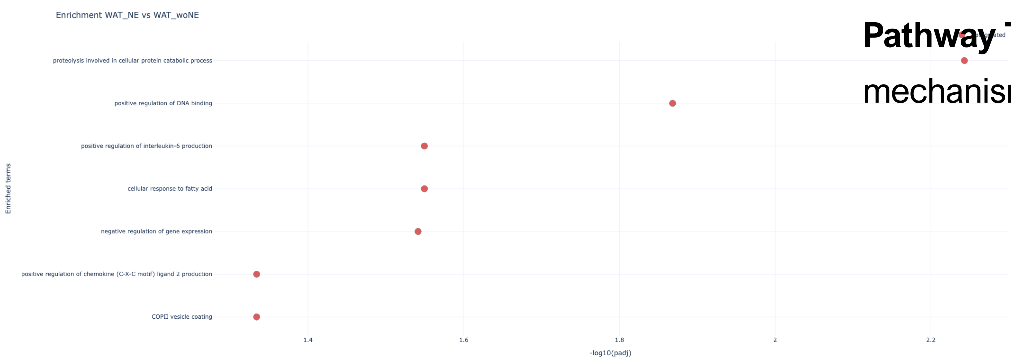


Functional Enrichment: Identify the biological functions, pathways, or processes associated with the differentially regulated proteins.

ORA (Over-Representation Analysis): Uses a hypergeometric test (Fisher's) to see if a "crowd" of your significant proteins belong to a specific pathway (e.g., Glycolysis).

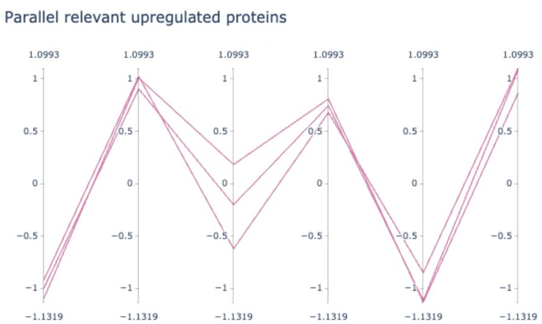
GSEA (Gene Set Enrichment Analysis): Considers the *entire* ranked list of proteins, not just a cutoff. It detects subtle but consistent shifts in biological processes.

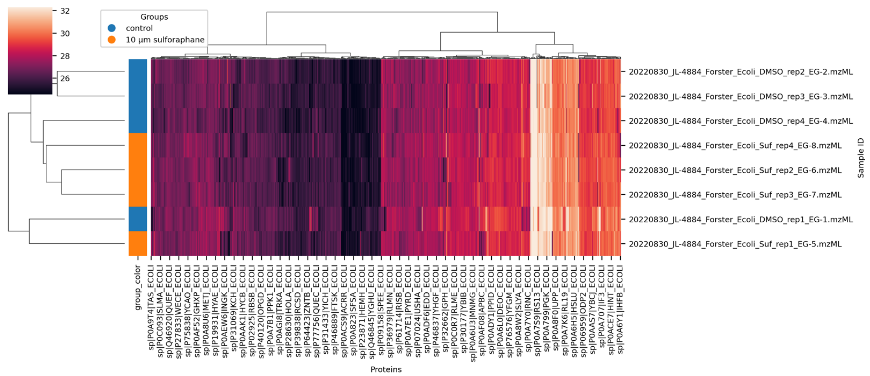
Pathway Tools: KEGG and **STRING** for mapping proteins to molecular mechanisms.



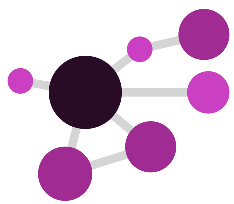
Other Potential Analysis

Clustering/Pattern Discovery hierarchical clustering of samples and proteins – heatmap, profile plots, correlation analysis.

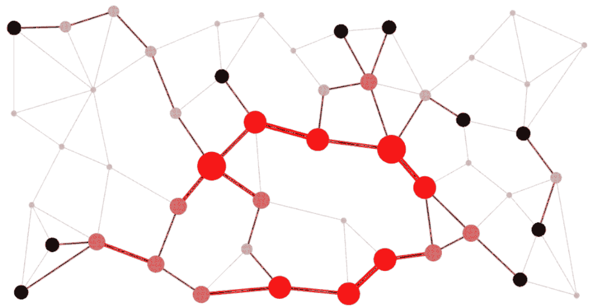




Graph Analysis: Build a protein graph/network and use the structure of nodes and relationships to find relevant patterns.



Graphs

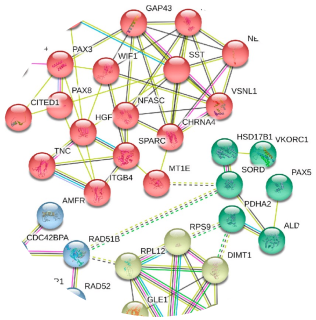


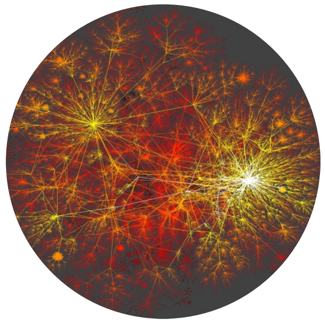
What is a Graph/Network?

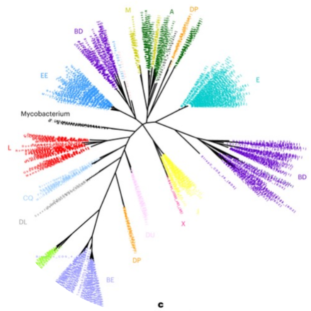
- Data structures of **components (nodes)** connected by **relationships (edges)**

Social networks

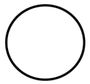
Biological networks





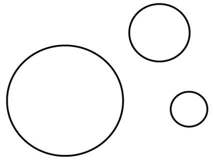


Graphs



Node

Edge

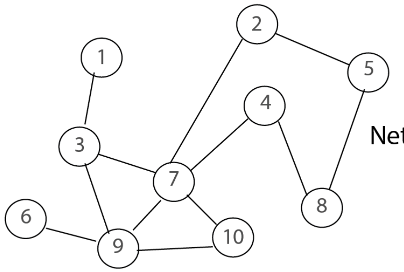


weighted nodes (size)

weighted edges (thickness)

undirected edge

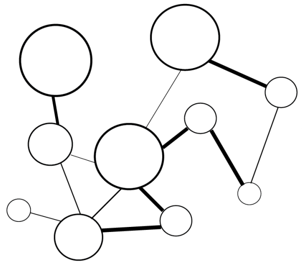
directed edge



Network

$$\begin{bmatrix} 0&0&1&0&0&0&0&0&0&0 \\ 0&0&0&0&1&0&1&0&0&0 \\ 1&0&0&0&0&0&1&0&1&0 \\ 0&0&0&0&0&0&1&1&0&0 \\ 0&1&0&0&0&0&0&1&0&0 \\ 0&0&0&0&0&0&0&0&1&0 \\ 0&1&1&1&0&0&0&0&1&1 \\ 0&0&0&1&1&0&0&0&0&0 \\ 0&0&1&0&0&1&1&0&0&1 \\ 0&0&0&0&0&0&1&0&1&0 \end{bmatrix}$$

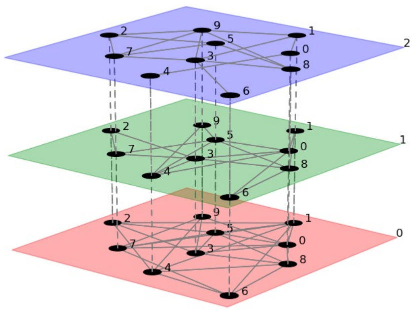
Adjacency matrix

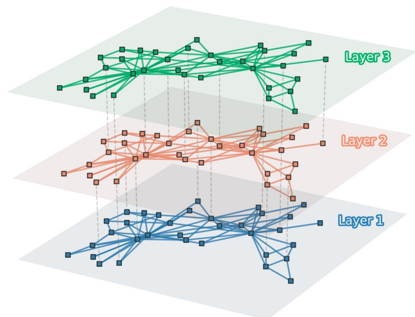


weighted undirected network (thickness)

$$\begin{bmatrix} 0&0&W_{1,3}&0&0&0&0&0&0&0 \\ 0&0&0&0&W_{2,5}&0&W_{2,7}&0&0&0 \\ W_{3,1}&0&0&0&0&0&W_{3,7}&0&W_{3,9}&0 \\ 0&0&0&0&0&0&W_{4,7}&W_{4,8}&0&0 \\ 0&W_{5,2}&0&0&0&0&0&W_{5,8}&0&0 \\ 0&0&0&0&0&0&0&0&W_{6,9}&0 \\ 0&W_{7,2}&W_{7,3}&W_{7,4}&0&0&0&0&W_{7,9}&W_{7,10} \\ 0&0&0&W_{8,4}&W_{8,5}&0&0&0&0&0 \\ 0&0&W_{9,3}&0&0&W_{9,6}&W_{9,7}&0&0&W_{9,10} \\ 0&0&0&0&0&0&W_{10,7}&0&W_{10,9}&0 \end{bmatrix}$$

Weighted adjacency matrix





Why Graphs?

- These structures allow:
  - Quick **integration** of **heterogeneous data** based on relationships
  - **Graph theory** methods can be used to **analyse** and **interpret** data, e.g., topological properties can be used to explain:
    - The possible **role** of specific components
    - The **flow** of information
    - The **robustness** of the system
- **Visualize** data

How to Analyse Graph Structures

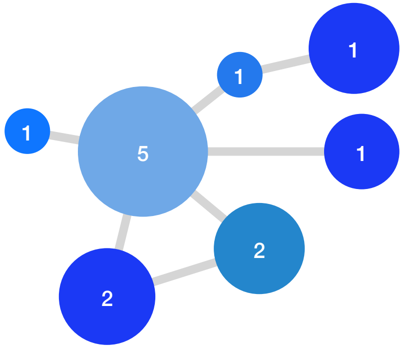
- **Graph Theory**: algorithms that allow you to extract relevant information from the topology of the graph.
  - **Topological Features:** Centrality, degree, clustering, etc.

- **Graph Machine Learning**:
  - Embeddings
  - Graph Neural Networks

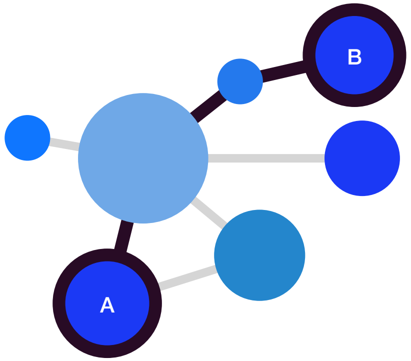
https://huggingface.co/blog/intro-graphml

Some Topological Features

Topological properties can help extract meaningful information and identify relevant structures within the network

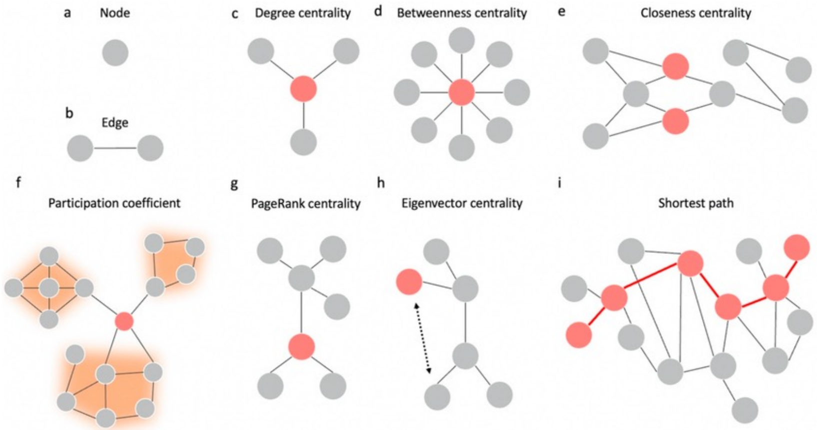


Degree

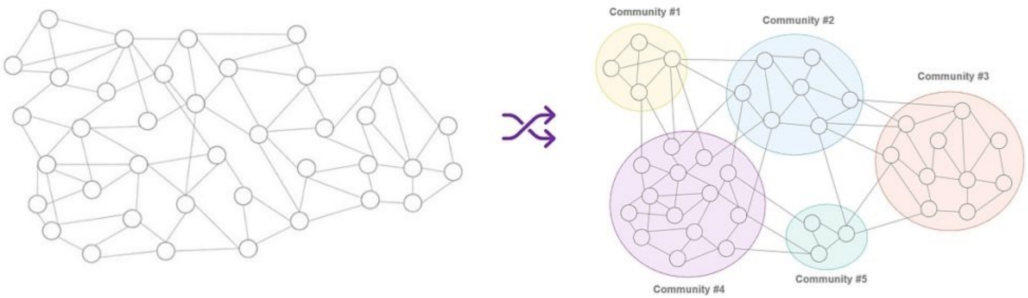


Shortest path

Centrality



Clustering



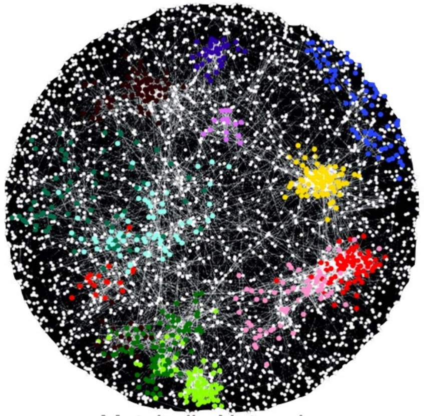
https://timbr.ai/community-detection-algorithm/

Graph Theory Measures and Their Application to Neurosurgical Eloquence. Cancers 2023

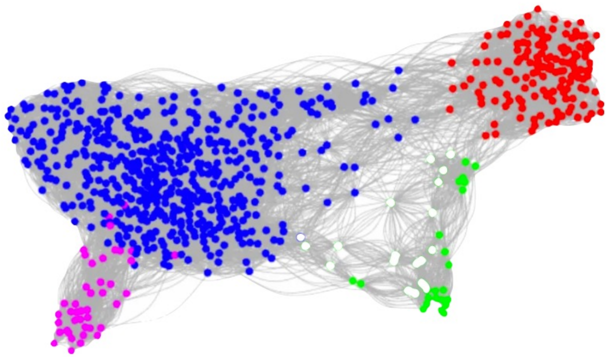
Graphs in Biology

https://towardsdatascience.com/umap-for-data-integration-50b5cfa4cdcd
http://snap.stanford.edu/deepnetbio-ismb/ipynb/Human+Disease+Network.html
https://cytoscape.org/cytoscape-tutorials/presentations/ppi-tools1-2017-mpi.html#/
https://en.wikipedia.org/wiki/Metabolic_network
https://www.scienceandfood.org/the-flavor-network/

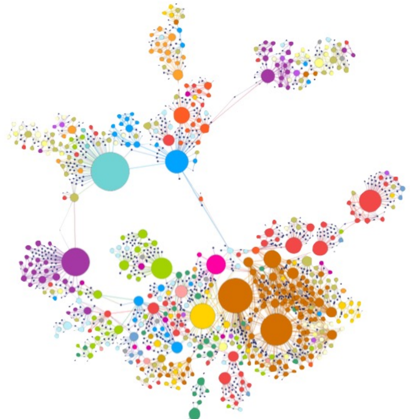
Protein-protein Interaction Networks



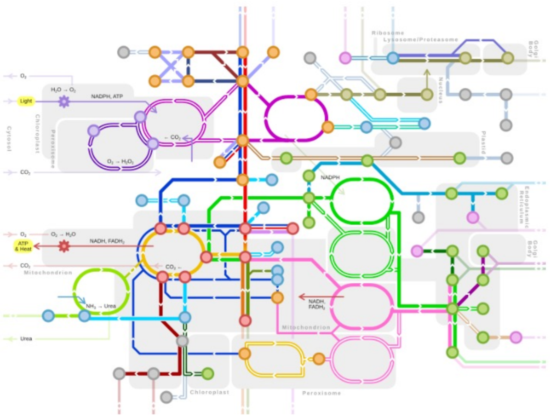
Single cell Networks



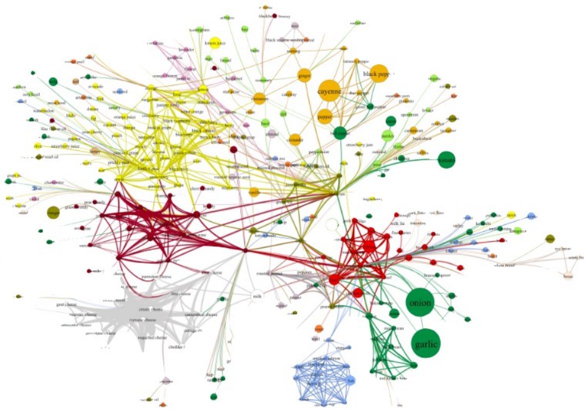
Disease Networks



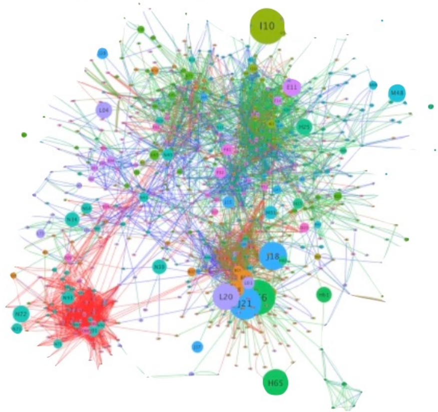
Metabolic Networks



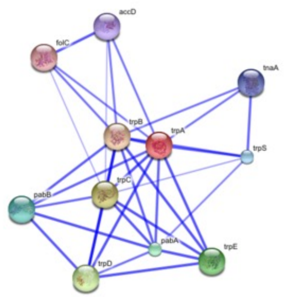
Food Networks

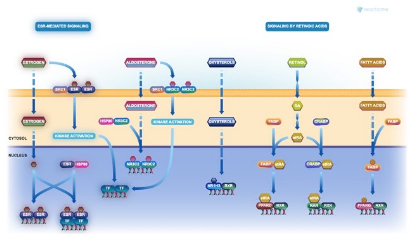


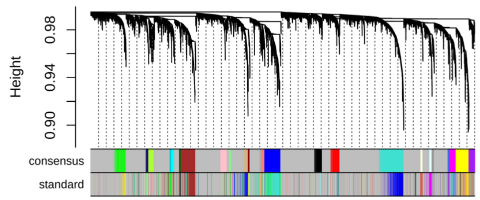
Diagnosis Progression Networks

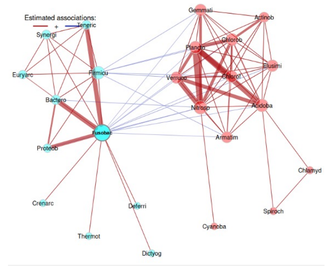


Data to Graph









Data sources

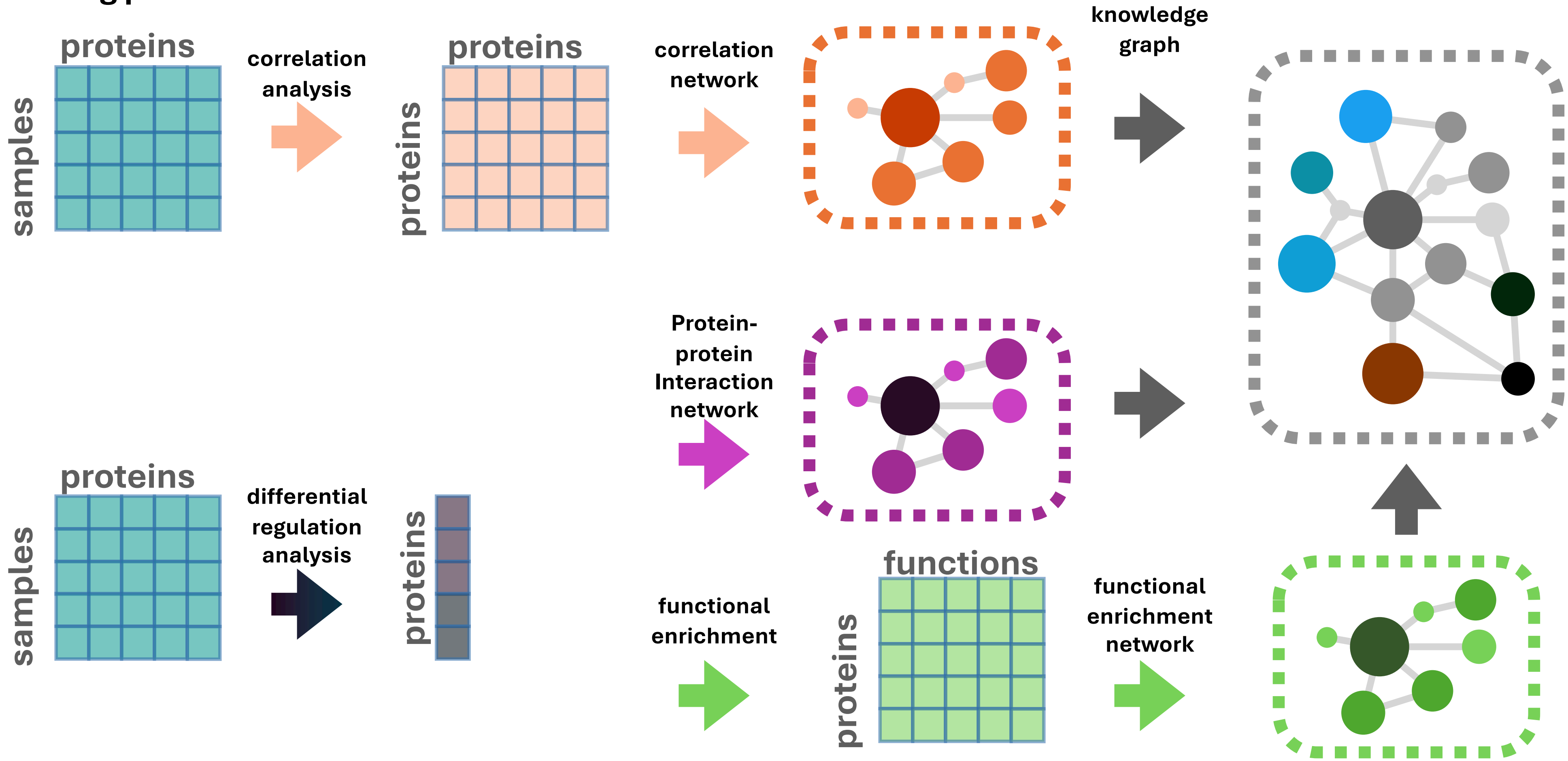
- STRING — https://string-db.org/
- BioGRID — https://thebiogrid.org/
- IntAct — https://www.ebi.ac.uk/intact
- REACTOME — https://reactome.org/
- KEGG — https://www.genome.jp/kegg/
- MINT — https://mint.bio.uniroma2.it/

Correlation-based networks — constructed by calculating pairwise correlations between entities based on their expression profiles across multiple conditions, time points, or samples (Weighted gene co-expression network analysis (WGCNA), co-abundance networks)

Knowledge-base approaches — also called knowledge graphs and built by integrating heterogeneous data from multiple sources —> <u>Knowledge Graphs</u>

How to Build a Network

Starting point



Knowledge Graphs

Knowledge Graphs

- A way to organise **knowledge/information** by defining associations or relationships
- These relationships facilitate **integration**, **management** and **enrichment** of data
- The objective when setting up a KG:

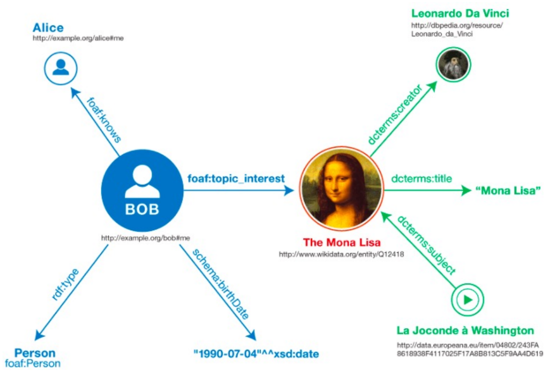
Standardisation / FAIRification

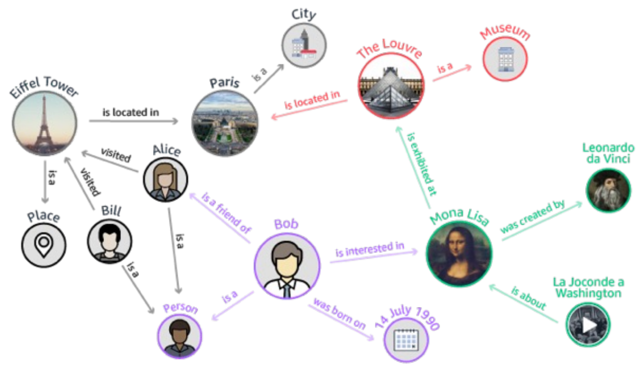
Reusability

Interpretability

Automation

Representation/Visualisation

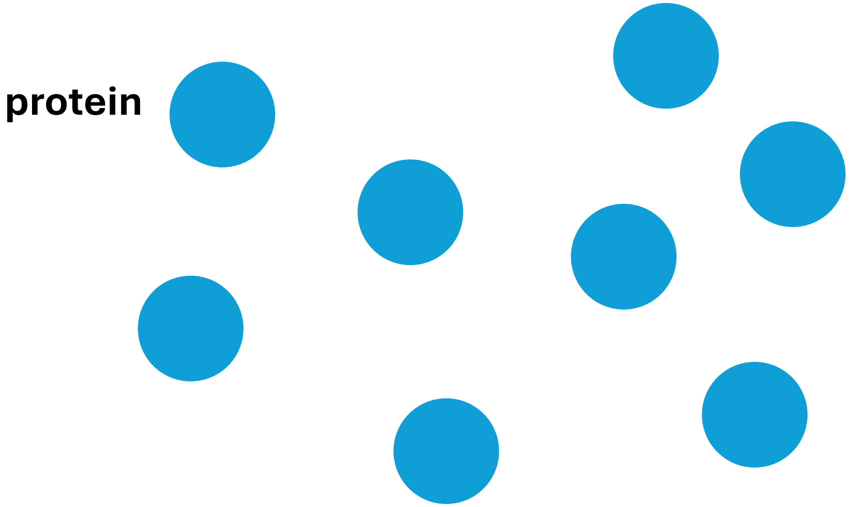




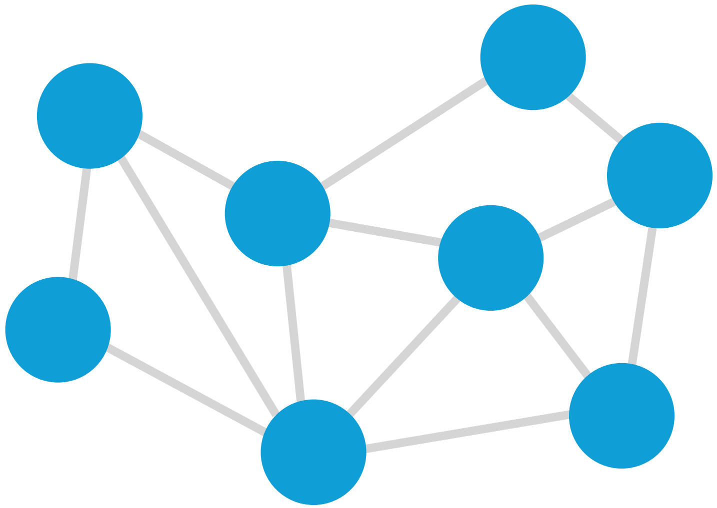
The Knowledge Graph Cookbook. Andreas Blumauer and Helmut Nagy. 2020

In proteomics?

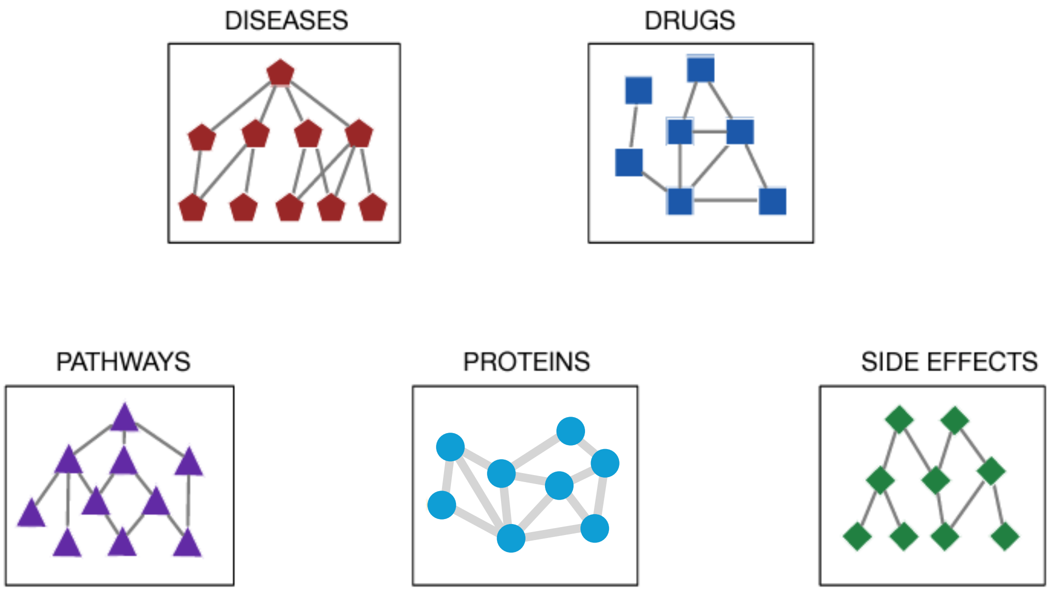
How Does It Work?



How Does It Work?



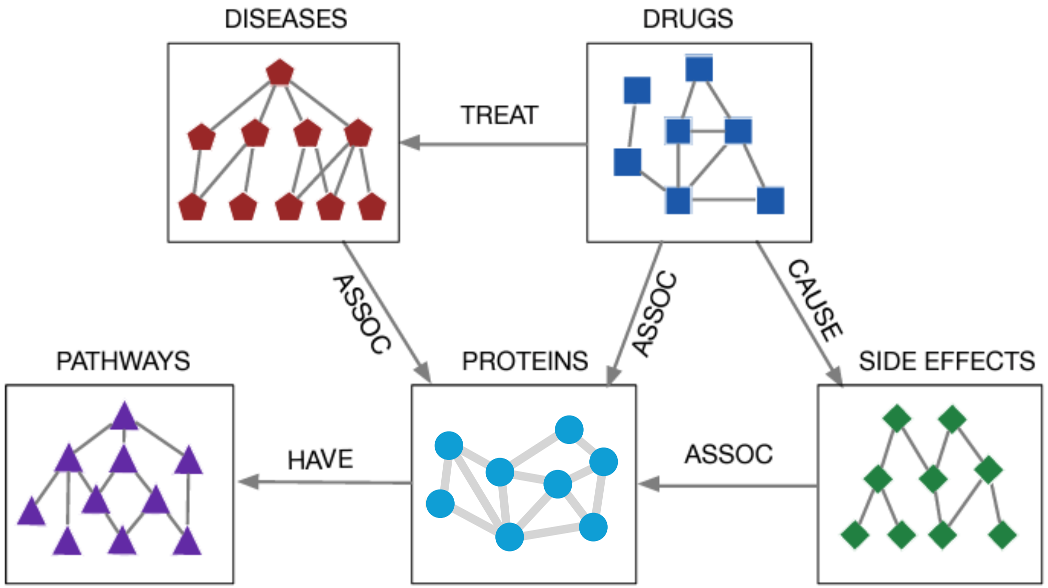
How Does It Work?

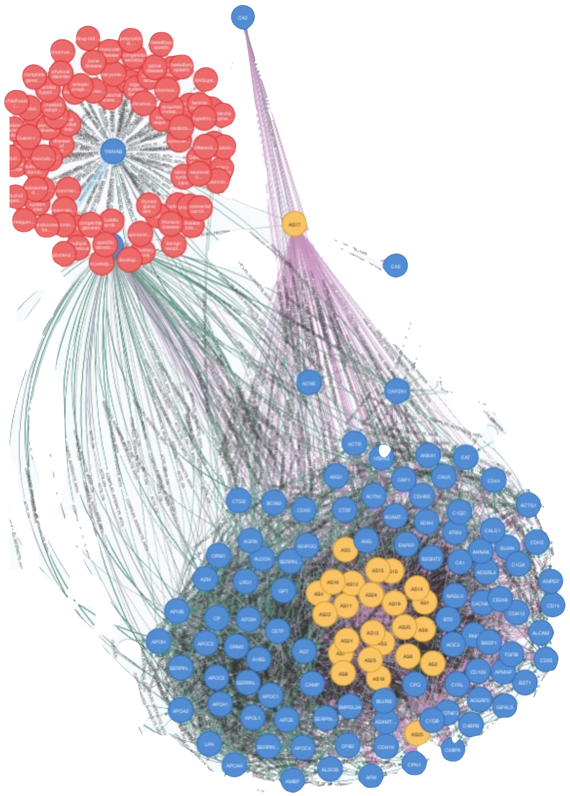


https://snap.stanford.edu/graphlearning-workshop/slides/stanford_graph_learning_Biomedicine.pdf
https://snap.stanford.edu/graphlearning-workshop/slides/stanford_graph_learning_Biomedicine.pdf

Knowledge Graph

Focus on data integration to represent complex biological systems and be able to reason over them

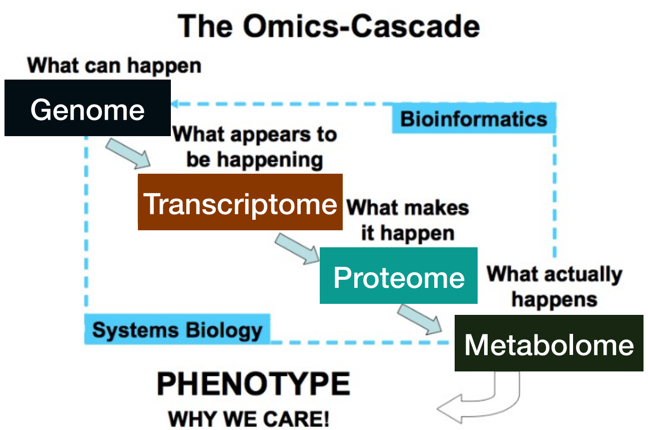


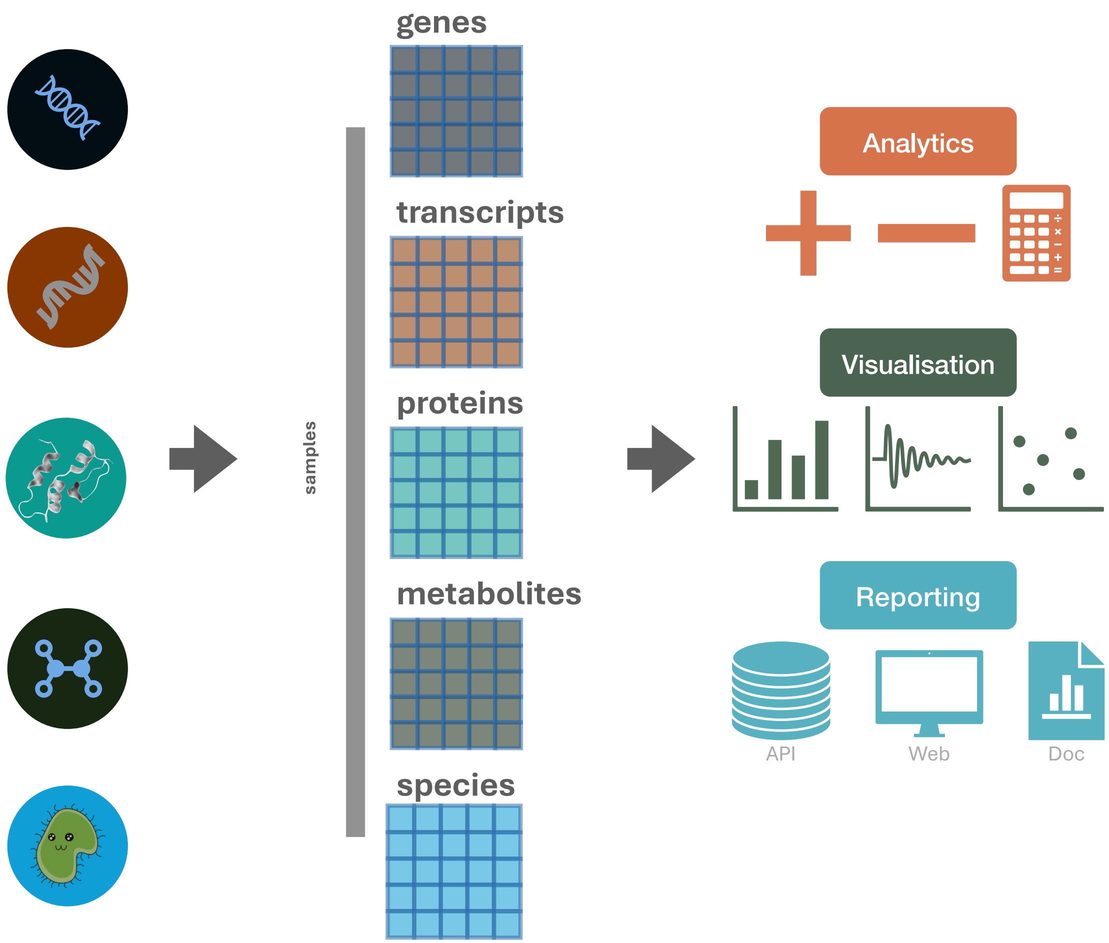


https://snap.stanford.edu/graphlearning-workshop/slides/stanford_graph_learning_Biomedicine.pdf
https://snap.stanford.edu/graphlearning-workshop/slides/stanford_graph_learning_Biomedicine.pdf

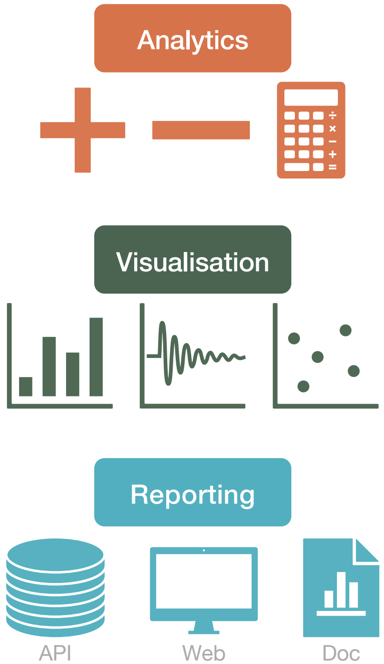
Open Source Tools

Omics Data


The Omics-Cascade
What can happen
Genome
Bioinformatics
What appears to be happening
Transcriptome
What makes it happen
Proteome
What actually happens
Systems Biology
Metabolome
PHENOTYPE
WHY WE CARE!


samples
genes
transcripts
proteins
metabolites
species
Analytics
Visualisation
Reporting
API
Web
Doc

MoNA Open Source Tools

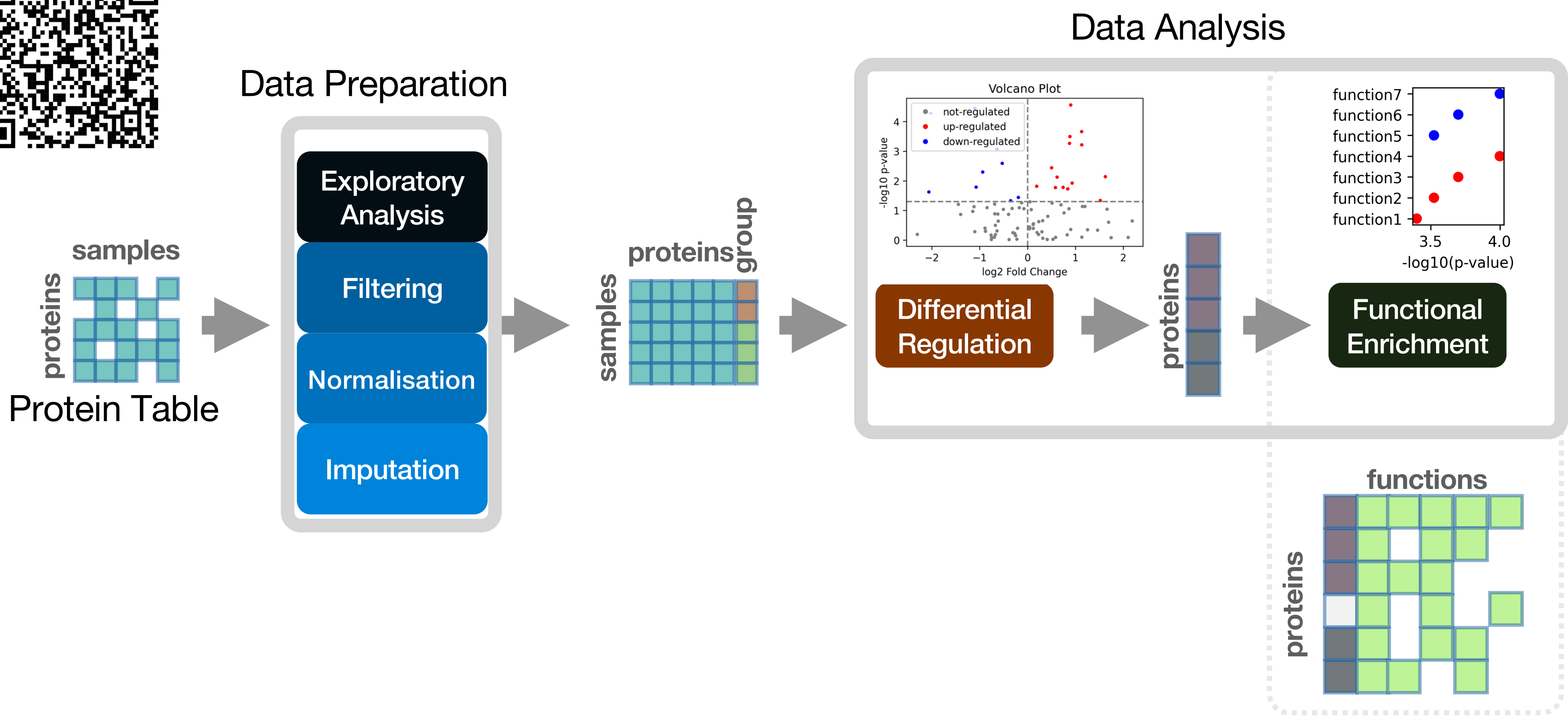


Analytics core library — analytics-core.readthedocs.io/

Visualization core library — github.com/Multiomics-Analytics-Group/vuecore

Automated Reporting library and cli — github.com/Multiomics-Analytics-Group/vuegen

Acore – Analytical core – workflow example



Other Tools for Downstream Proteomics Analysis

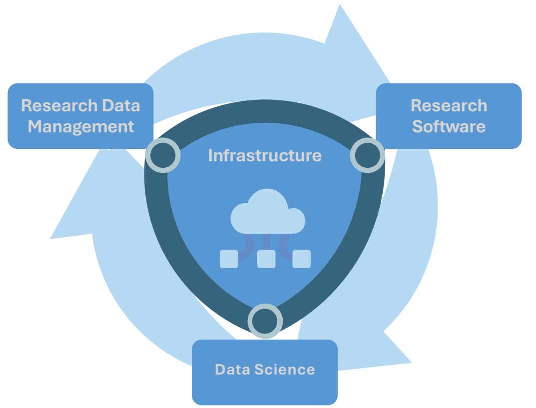
Category	Tool	
Statistical Analysis	Perseus, limma, MSstats, AlphaStats	GUI and R/Python-based options
Functional Enrichment	Enrichr	Web tool
Pathway Analysis	Reactome, IPA (Qiagen)	Curated databases
Network Analysis	STRING, Cytoscape, Gephi	Visual and analytical network tools
Integrated Platforms	CKG, Proteome Discoverer, AlphaPept	Combine multiple steps

Acknowledgements

DTU

Multi-omics Network Analytics Research Group

Informatics Platform



BRiGHT

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https://github.com/Multiomics-Analytics-Group

https://multiomics-analytics-group.github.io/